

Max Dupré la Tour

Postdoctoral Researcher · RIKEN AIP and University of Tokyo

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Research Profile

I work on algorithmic game theory, fair division, graph theory, and approximation algorithms. I am a postdoctoral researcher in the RIKEN AIP Computational Social Choice Team and at the University of Tokyo.

Appointments

RIKEN Center for Advanced Intelligence Project <i>Postdoctoral Researcher, Computational Social Choice Team</i>	2026–present Tokyo, Japan
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Education

McGill University <i>Ph.D. in Computer Science · Advisor: Adrian Vetta</i>	2022–2026 Montréal, Canada
Université Paris Diderot <i>M.Sc. in Mathematical Logic and Foundations of Computer Science</i>	2018–2019 Paris, France
École Normale Supérieure de Paris <i>M.Sc., Department of Mathematics and Applications</i>	2015–2019 Paris, France
Université Pierre et Marie Curie <i>B.Sc. in Mathematics</i>	2013–2015 Paris, France

Publications

Balanced Fair Division for Three Agents under General Valuations and Laminar Constraints Preprint · arXiv	2026
From Cake-Cutting and Necklace-Splitting to Fair Division of Indivisible Items with Ayumi Igarashi · Preprint · arXiv	2026
Bad News for Couples: Bounds for Fair Division of Indivisible Items among Groups Preprint · arXiv	2026
Discrepancy and Fair Division for Non-Additive Valuations with Kaito Fujii · Preprint · arXiv	2025
Random Serial Dictatorship is $\sqrt{2}$-Envy-Free with Frank Connor, Louis-Roy Langevin, Vishnu V. Narayan, Ndiamé Ndiaye, Neil Rahman, and Adrian Vetta · <i>EC 2026</i> · <i>Exemplary Paper</i> · arXiv	2026
Tight Asymptotic Bounds for Fair Division With Externalities with Frank Connor, Vishnu V. Narayan, and Šimon Schierreich · <i>IJCAI 2026</i> · arXiv	2026
Faster and Simpler Greedy Algorithm for k-Median and k-Means with David Saulpic · <i>ICALP 2026</i> · arXiv	2026
On the Hardness of Recognizing Graphs of Small mim-width and Its Variants with Manuel Lafond and Ndiamé Ndiaye · <i>ICALP 2026</i> · arXiv	2026
Recognizing Leaf Powers and Pairwise Compatibility Graphs is NP-Complete with Manuel Lafond and Ndiamé Ndiaye · <i>SODA 2026</i> · arXiv	2026
k-Leaf Powers Cannot Be Characterized by a Finite Set of Forbidden Induced Subgraphs with Manuel Lafond, Ndiamé Ndiaye, and Adrian Vetta · <i>ICALP 2025</i> · arXiv	2025
Eliminating Majority Illusion is Easy with Jack Dippel, April Niu, Sanjukta Roy, and Adrian Vetta · <i>AAAI 2025</i> · <i>Best Poster</i> · arXiv	2025
Making Old Things New: A Unified Algorithm for Differentially Private Clustering with David Saulpic and Monika Henzinger · <i>ICML 2024</i> · <i>Spotlight</i> · arXiv	2024
Fully Dynamic Euclidean k-Means Coreset in Constant Update Time with David Saulpic and Monika Henzinger · <i>ESA 2024</i>	2024
Gerrymandering Planar Graphs	2024

with Jack Dippel, April Niu, Sanjukta Roy, and Adrian Vetta · *AAMAS 2024*

Mitigating COVID-19 Outbreaks in Workplaces and Schools by Hybrid Telecommuting 2021

with Simon Mauras, Vincent Cohen-Addad, Guillaume Duboc, Paolo Frasca, Claire Mathieu, Lulla Opatowski, and Laurent Viennot · *PLOS Computational Biology* 17(8)

The Combinatorial Clock Auction: The Effects of Strategic Behaviour and the Price Increment Rule on Social Welfare 2018

with Adrian Vetta · *EC 2018* · DOI

Research Internships

National Institute of Informatics 2025 · 5 months
Discrepancy and fair division · Supervisor: Kaito Fujii
Tokyo, Japan

Institute of Science and Technology Austria 2023 · 2 months
Differential privacy and clustering · Supervisor: Monika Henzinger
Klosterneuburg, Austria

Université Paris Diderot 2019 · 4 months
Circuit complexity · Supervisor: Sylvain Perifel
Paris, France

McGill University 2018 · 4 months
Algorithmic game theory and spectrum auctions · Supervisor: Adrian Vetta
Montréal, Canada

Teaching Experience

Teaching Assistant 2020–2024
Algorithms, discrete mathematics, and probability

Mathematics Tutor 2020–2021
High-school and undergraduate students

Mathematics Test-Preparation Instructor 2016–2017
Advanced preparation for the French Grandes Écoles

Academic Service

Program committee: WINE 2026; AAAI 2026.

Skills

Programming Python, L^AT_EX
Languages French (native), English (proficient)